

SUBHRAJYOTY ROY

📍 **Nationality:** Indian
☎️ (+1) (314)745-2968
✉️ roys@wustl.edu

🎓 [Google Scholar](#)
🔗 github.com/subroy13
🌐 [/in/subroy13](#)
🌐 [Portfolio Website](#)

CURRENT POSITION

- *Postdoctoral researcher* at the Department of Statistics & Data Science in the **Washington University, St. Louis**.

RESEARCH INTERESTS

- Robust statistical inference, divergences, minimum divergence estimation.
- Dependent data analysis, spatial analysis, spatiotemporal analysis.
- Statistical methods for Deep learning, AI safety, and trustworthiness.

EDUCATION

2021 - 2025	Ph.D. in Statistics Thesis: Robust Matrix Factorization using the Density Power Divergence and its Applications. Supervisors: Prof. Ayanendranath Basu and Prof. Abhik Ghosh.	Indian Statistical Institute, Kolkata
2019-2021	Masters of Statistics (M.Stat.) Grade: 93.8%, Financial Statistics specialization, Ranked <i>1st</i> .	Indian Statistical Institute, Kolkata
2016 - 2019	Bachelor of Statistics (B.Stat.) Grade: 93.6%, Ranked <i>2nd</i> .	Indian Statistical Institute, Kolkata
2014 - 2016	Higher Secondary (WBCHSE) Grade: 93.4%, District Rank <i>2nd</i> .	Nimta High School, Kolkata
2014	Secondary (WBCSSE) Grade: 94.57%, State Rank <i>20th</i> .	Nimta High School, Kolkata

PROFESSIONAL EXPERIENCE

July, 2025-	Postdoctoral Researcher Department of Statistics and Data Science, Washington University at St. Louis Research Mentor: Prof. Debashis Mondal.	
July, 2022 - June, 2025	Principal Information Researcher • Continuous system improvement, research, and novel technology development across the company. • Implementing a centralized control flow system to reduce resource overconsumption, resulting in 90% reduction in API rate-limit errors. • Development of SysCloud Sky and Scout, the AI-powered chatbots for customer and internal usage. • Building company-wide automation and adoption to customized AI solutions using prompting, RAG, AI agents, etc.	SysCloud Inc.
May, 2024 - July, 2024	Mentor • Course instructor for <i>Mathematics for Artificial Intelligence</i> module. Received a consistent rating of 4.5+.	Great Learning Education Series Pvt. Ltd.

July, 2021 - June, 2022	Data Scientist	SysCloud Inc.
	- Developed models for optimized bidding strategy for Google Ads using reinforcement learning. - Unified, scalable and robust system architecture development of backup, restore and export of any cloud, resulting in business expansion for various SaaS apps (Quickbooks, Hubspot, Salesforce, Slack, Box, etc.).	
2020 May - June	Data Scientist Intern	General Electric (Aviation)
	Understanding of the impact of COVID-19 on the aviation industry with an analysis of historical tweets data.	
2019 May - June	Data Scientist Intern	SysCloud Inc.
	- Developed cost optimization techniques for backup and restore of G-suite and Office 365 workspaces. - Solving phishing email detection problem using a combination of NLP and Deep learning.	

PUBLICATIONS

Journal Articles

1. **Roy, S.**, Sarkar, A., Ghosh, A., & Basu, A. (2026). Asymptotic Breakdown Point Analysis for a General Class of Minimum Divergence Estimators. *Bernoulli*, 32(1), 698-722. [\[Link\]](#).
2. **Roy, S.**, Basu, A., & Ghosh, A. (2024). Robust Principal Component Analysis using Density Power Divergence. *Journal of Machine Learning Research*, 25(324), 1-40. [\[Link\]](#).
3. **Roy, S.**, Ghosh, A., & Basu, A. (2024). Robust singular value decomposition with application to video surveillance background modelling. *Statistics and Computing*, 34(5), 178. [\[Link\]](#).
4. Pyne, A., **Roy, S.**, Ghosh, A., & Basu, A. (2024). Robust and efficient estimation in ordinal response models using the density power divergence. *Statistics*, 58(3), 481-520. [\[Link\]](#).
5. Bhaduri, R., **Roy, S.**, & Pal, S. K. (2022). Rough-Fuzzy CPD: a gradual change point detection algorithm. *Journal of Data, Information and Management*, 4(3), 243-266. [\[Link\]](#).
6. Ghatak, A., Singh Patel, S., Bonnerjee, S., & **Roy, S.** (2022). A generalized epidemiological model with dynamic and asymptomatic population. *Statistical Methods in Medical Research*, 31(11), 2137-2163. [\[Link\]](#).
7. **Roy, S.**, Sengupta, D., Rudra, K., & Saha, U. S. (2020). Analysis of Pollution Patterns in Regions of Kolkata. *Calcutta Statistical Association Bulletin*, 72(2), 133-170. [\[Link\]](#).
8. Mukherjee, D., Dalal, A., & **Roy, S.** (2019). Feasibility of Transparent Price Discovery in Tea through Auction in India. *Commodity Insights Yearbook (MCX)*, 44-52. [\[Link\]](#).

Preprints

1. **Roy, S.**, Ghosh, A., & Basu, A. (2025). Robust Rank Estimation for Noisy Matrices. *arXiv preprint arXiv:2510.19583*. [\[Link\]](#).
2. Bonnerjee, S., Karmakar, S., & **Roy, S.** (2025). WISER: Segmenting watermarked region-an epidemic change-point perspective. *arXiv preprint arXiv:2509.21160*. [\[Link\]](#).
3. Jana, S., **Roy, S.**, Ghosh, A., & Basu, A. (2025). Asymptotic breakdown point analysis of the minimum density power divergence estimator under independent non-homogeneous setups. *arXiv preprint arXiv:2508.12426*. [\[Link\]](#).

4. **Roy, S.**, Basu, S., Ghosh, A., & Basu, A. (2025). Characterization of Generalized Alpha-Beta Divergence and Associated Entropy Measures. *arXiv preprint* arXiv:2507.04637. [\[Link\]](#).
5. Deb, S., Neves, C., & **Roy, S.** (2024). Nonparametric quantile regression for spatio-temporal processes. *arXiv preprint* arXiv:2405.13783. [\[Link\]](#).
6. **Roy, S.** (2024). Trustworthy Dimensionality Reduction. *arXiv preprint* arXiv:2405.05868. [\[Link\]](#).
7. Bhaduri, R., Bonnerjee, S., & **Roy, S.** (2019). Onset detection: A new approach to QBH system. *arXiv preprint* arXiv:1908.07409. [\[Link\]](#).

SEMINARS, CONFERENCES & LECTURES

1. Invited lecture for **Statistics & Data Seminar Series, 2026** at the **Mathematics Department, Saint Louis University**.
Title: Localized Detection of Authenticity of Mixed LLM-Human Generated Texts.
2. Invited lecture for **Graduate Student Seminar Series, 2025** at the **Department of Statistics and Data Science, Washington University in St. Louis**.
Title: Localized Detection of Authenticity in Mixed Source Texts via Epidemic Change-point Perspective.
3. Student Paper Competition at **IISA Conference, 2024** at **Cochin University of Science and Technology**.
Title: Robust Principal Component Analysis using Density Power Divergence.
4. Contributory talk at **IMS Asia Pacific Rim Meeting, 2024** at **University of Melbourne**.
Title: A novel and scalable background modelling algorithm from video surveillance data.
5. Prasanta Chandra Mahalanobis memorial lecture at **Indian Statistical Institute, Kolkata**.
Title: Algorithmic Fairness of Statistical Decision Systems.
6. D. Basu memorial lecture at **Indian Statistical Institute, Kolkata**.
Title: tSNE: A way to visualize multivariate data.

TEACHING ACTIVITIES

Fall, 22	Robust Statistics Invited Guest Lecture on "A review of Robust Location and Scatter estimators."	Indian Statistical Institute, Kolkata
Summer, 24	Mathematics for Artificial Intelligence	Great Learning Education Series Pvt. Ltd.
Fall, 25	SDS 4010 Introduction to Probability	Washington University in St. Louis
Spring, 26	SDS 5140 Advanced Linear Models	Washington University in St. Louis

ADDITIONAL ACTIVITIES

- Maintain a substack newsletter [StatWizard](#) with 200+ subscribers.
- Served as a session chair for IMS Asia-Pacific Rim Meeting 2024 at the University of Melbourne, Australia.
- Reviewer for the journal **Scientific Reports**.

PROJECTS, SOFTWARE AND PACKAGES

2025	handanim - Contributor & Maintainer A Python library to create whiteboard-style, hand-drawn animations for educational videos, tutorials or data storytelling.	Github Link
2024	Langchain - Contributor Framework for developing applications powered by large language models	Github Commits
2023	Multi PDF QueryBot - Developer & Maintainer A streamlit application to chat with PDF documents using RAG and Large language models. Uses a locally hosted Chroma vector database, SQLite for metadata storage, and Claude for access to foundational LLM.	Github Link
2023	callgrid-reader - Developer & Maintainer A typescript utility package to read, parse and analyze callgrind files for profiling analysis of codes	NPM Link
2023	decompy - Developer & Maintainer Python package for multiple robust matrix and tensor decomposition algorithms	PyPI Link
2022	roufcp - Developer & Maintainer Python package implementing a gradual changepoint detection algorithm using rough-fuzzy set theory	PyPI Link
2021	rsvddpd - Developer & Maintainer R package to perform Robust SVD using Density Power Divergence	CRAN Link
2020	COVID-19 Tracker: A Dashboard for prediction and control assistance Developed a dashboard to visualize the COVID-19 situation in different states of India. It dynamically estimates the time-varying reproduction rate and prediction of COVID-19 spread for various lockdown scenarios for district levels in India, based on a modified eSIR model. The project was funded by the Indian Insititute of Management, Vishakhapatnam.	
2019	Analysis of Climber's challenges in scaling Mount Rainier Modelling of climber's success rate in climbing Mount Rainier based on weather modelling. Modelling of climber's team sizes for insights about mountaineering-based local economy.	Report Link
2019	An Introduction of Causal Inference using Direct Acyclic Graph Understanding of the setup and assumptions for causal inference. Estimating the effect of intervention using frontdoor and backdoor criterion, simulations and applications to real-world datasets to estimate causality.	Report Link
2018	Chatbot using Seq2Seq Attention Mechanism It is a chatbot created with a seq2seq neural network with a basic attention mechanism, trained with the Cornell Movie Corpus dataset.	Github Link

ACCOLADES

2024	Nominated for the Student Paper Award	IISA Conference 2024
2024	AWS Cloud Practitioner Certification	Amazon Web Services (AWS)
2023	AIR 6 at National Eligibility Test Council of Scientific and Industrial Research (CSIR), Govt. of India	
2022	Nominated for P. C. Mahalanobis Memorial Gold Medal Indian Statistical Institute, Kolkata	
2022	J. K. Ghosh Memorial Gold Medal	Indian Statistical Institute, Kolkata
2016-2021	Dean's list of toppers	Indian Statistical Institute, Kolkata
2020	Scaled score 324 at GRE General Test	Educational Testing Service (ETS)

2020	Nikhilesh Bhattacharya Memorial Gold Medal	Indian Statistical Insititute, Kolkata
2020	D. Basu Memorial Gold Medal	Indian Statistical Insititute, Kolkata
2019	93 percentile at GRE Mathematics Subject Test	Educational Testing Service (ETS)
2018	1st Quartile at Simon Marais Mathematical Competition Australian Mathematical Science Institute	
2016	AIR 1 in Undergraduate Entrance Exam Institute of Mathematics and Application, Bhubaneshwar	

SKILLS

Programming R, Python, SQL, PHP, JavaScript (NestJS, NodeJS, React, NextJS).
Technologies AWS (S3, Lambda, ECS, Sagemaker, Bedrock), Docker, Langchain, Langflow.
Tools MS-Office, $\LaTeX$

I certify that the information provided above is true and accurate to the best of my knowledge¹

— Subhrajyoty Roy

¹Last updated on Feb, 2026